

Guglielmo Cappellini

Postdoctoral Researcher in AI for Health

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Personal Details

- **Date of Birth:** 12 February 1996
- **Nationality:** Italian
- **Dutch Work Permit:** Yes, EU/EEA Citizen
- **Gender:** Male

Research Statement / Professional Summary

Curious and passionate researcher with a strong commitment to collaborative and interdisciplinary research for healthcare applications. Expert in Python and PyTorch for computational modelling and multimodal deep learning. Experienced in clear communication of complex technical concepts across deep learning, medicine, and mathematics. Autonomous mindset, with the objective to understand the clinical needs and how to develop innovative AI tools that have real-world impact.

Education

2022 - 2024 PhD in AI for Healthcare

Dept. of Computer, Automation, and Management Engineering, Sapienza Università di Roma, Rome, Italy

- "AI-based solution methods for PDEs with application to oncological hyperthermia"
- Supervisor: Prof. M. Vendittelli
- Travel Award and Two-papers work presented at IEEE CDC 2023, Singapore

2018 - 2021 MSc in Nuclear Engineering

Dept. of Aerospace, Electrical, and Energy Engineering, Sapienza Università di Roma, Rome, Italy

- Thesis: "Study and characterization of radiolabeled nanovectors for application in precision molecular medicine", at Dept. of Nuclear Medicine Policlinico Umberto I, Roma

2015 - 2017 BSc in Energy and Nuclear Engineering

Dept. of Aerospace, Electrical, and Energy Engineering, Sapienza Università di Roma, Rome, Italy

- Thesis: "Radionuclides production and medical applications"

Academic Appointments / Professional Experience

2025 - present Postdoctoral Researcher

Dept. of Computer, Automation, and Management Engineering, Sapienza Università di Roma, Rome, Italy

- Developing deep learning models (CNNs, geometric deep learning, and PINNs) for real-time simulation of deformable tissues with application to interactive virtual reality for medical training
- Advancing the research on AI-based tools (PINNs, RNNs) for non-invasive temperature estimation of oncological hyperthermia treatments

2024 Visiting PhD student (6 months)

Dept. of Radiation Oncology, Amsterdam UMC Hospital, Amsterdam, the Netherlands

- Designed and implemented the experimental validation of PINNs for non-invasive temperature estimation in superficial hyperthermia treatments

Research Output

Journal Articles:

1. **Cappellini, G.**, Cristofaro, A., De Santis, E., Staffetti, E., Trappolini, G., Vendittelli, M. "Adaptive Estimation of Pennes' Bio-Heat Equation: Observer Design and PINNs-based Implementation", *Under review at IEEE Transactions on Control Systems Technology*.

Conference Proceedings (with full papers, peer-reviewed):

1. Cristofaro, A., **Cappellini, G.**, Staffetti, E., Trappolini, G., Vendittelli, M., (2023). "Adaptive estimation of the Pennes' Bio-Heat equation – I: Observer design." Proceedings of the 62nd IEEE Conference on Decision and Control, Singapore, 1931-1936.
2. **Cappellini, G.**, Trappolini, G., Staffetti, E., Cristofaro, A., Vendittelli, M., (2023). "Adaptive estimation of the Pennes' Bio-Heat equation – II: A NN-Based Implementation for Real-Time Applications." Proceedings of the 62nd IEEE Conference on Decision and Control, Singapore, 5364-5369.
3. **Cappellini, G.**, Cristofaro, A., Staffetti, E., Trappolini, G., Vendittelli, M., (2023). "Adaptive estimation of Pennes' Bio-Heat Equation." Proceedings of 2023 I-RIM Conference, Rome, Italy, 202-206.

Selected Conference Presentations:

- 2023, "Adaptive estimation of the Pennes' Bio-Heat equation – II: A NN-Based Implementation for Real-Time Applications." Oral presentation, 62nd IEEE Conference on Decision and Control, Singapore.
- 2023, "Adaptive estimation of the Pennes' Bio-Heat equation." Oral and poster presentation, 2023 I-RIM Conference, Rome, Italy.

Teaching Experience and Philosophy

- **2022/2023 & 2024/2025 Fundamentals of Automation**, Dept. of Information, Electronics, and Telecommunications Engineering, Sapienza Università di Roma, Role: Tutor
 - Responsibilities: Designed syllabus, supervised BSc student projects and examinations
- **2022/2023 Medical Robotics**, Dept. of Computer, Automation, and Management Engineering, Sapienza Università di Roma, Role: Final project supervisor
 - Supervised 5 MSc students teams for the final project

Teaching Philosophy

"To foster critical thinking and active learning through discussion and hands-on sessions. I am dedicated to encourage responsible use of LLMs and AI, leveraging technology to enhance student engagement and understanding of complex concepts in AI for Healthcare."

Supervision Experience

- **Master's Students:** 2 completed
- **Bachelor's Students:** 1 completed

Academic Service

- **Reviewer for Conference:** 2023 I-RIM Conference, Rome, Italy

Professional Memberships

- IEEE, IEEE Young Professionals, and IEEE Control Systems Society, 2023-2024

Skills

- Deep Learning & AI: Expert in developing and implementing multimodal deep learning models (CNNs, ViTs, geometric deep learning, PINNs, RNNs) using Python and PyTorch; experienced in computer vision and medical image analysis
- Medical Data Integration: experienced in ontology-based data preparation and analysis of large-scale healthcare datasets,
- Computational Modeling: PINNs for PDEs, computational methods (MATLAB), HPC
- Interdisciplinary Collaboration: Proven experience working with clinical partners, medical professionals, and industrial stakeholders in healthcare innovation projects
- Languages: Italian (native), English (fluent, IELTS Academic: 7.5), Spanish
- Professional Qualification: Licensed Professional Industrial Engineer (Esame di Stato Sez. A)

Awards and Distinctions

- 2023 Travel support IEEE CDC23, IEEE

References

- Prof. M. Vendittelli
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- Dr. H. Crezee
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